some of their key colleagues: Wheeler with Oppenheimer; Zel'dovich with Landau and with Andrei Sakharov.

John Wheeler, fresh out of graduate school in 1933, and the winner of a Rockefeller-financed National Research Council postdoctoral fellowship, had a choice of where and with whom to do his postdoctoral study. He could have chosen Berkeley and Oppenheimer, as did most NRC theoretical physics postdocs in those days; instead he chose New York University and Gregory Breit. "In personality they [Oppenheimer and Breit] were utterly different," Wheeler says. "Oppenheimer saw things in black and white and was a quick decider. Breit worked in shades of grey. Attracted to issues that require long reflection, I chose Breit."

From New York University in 1933, Wheeler moved on to Copenhagen to study with Niels Bohr, then to an assistant professorship at the University of North Carolina, followed by one at Princeton University, in New Jersey. In 1939, while Oppenheimer and students in California were probing neutron stars and black holes, Wheeler and Bohr at Princeton (where Bohr was visiting) were developing the theory of *nuclear fission:* the breakup of heavy atomic nuclei such as uranium into smaller pieces, when the nuclei are bombarded by neutrons (Box 6.1). Fission had just been discovered quite unexpectedly by Otto Hahn and Fritz Strassman in Germany, and its implications were ominous: By a chain reaction of fissions a weapon of unprecedented power might be made. But Bohr and Wheeler did not concern themselves with chain reactions or weapons; they just wanted to understand how fission comes about. What is the underlying mechanism? How do the laws of physics produce it?

Box 6.1

Fusion, Fission, and Chain Reactions

The *fusion* of very light nuclei to form medium-sized nuclei releases huge amounts of energy. A simple example from Box 5.3 is the fusion of a deuterium nucleus ("heavy hydrogen," with one proton and one neutron) and an ordinary hydrogen nucleus (a single proton) to form a helium-3